

A Comprehensive Analysis of Cybersecurity Challenges in Self-Adaptive Avionics: A Plug&Fly Avionics Platform Case Study

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Abstract— Increasing complexity and availability demands of traditional software systems have derived the need for self-adaptive systems. Self-adaptive systems automatically adjust their behavior at runtime to meet objectives. However, these systems accompany additional challenges specially in terms of security. These challenges include interaction with non-trusted entities, widening of attack surface due to multichannel communication, risk of unauthorized and malicious access to operating environment, and potential exploitation of adaptive mechanisms by attackers. Despite the growing adoption of these systems in avionics, comprehensive study on their cybersecurity assessment is missing. To fill the gap, this study discusses various security vulnerabilities introduced with self-adaptivity in avionics. The aim of the study is to use the Plug&Fly Avionics Platform (PAFA) as a case study, and analyze the impact of self-adaptivity in exposing the system to a wide range of cyber threats such as, faulty decision making, compromise reconfigurations, and so on. Besides detailed analysis of security vulnerabilities introduced with self-adaptivity, the paper also proposes some novel mitigation strategies and demonstrate their effectiveness in achieving a cybersecure adaptive system in the context of PAFA. These findings aim to provide solid basis in future research and development towards securing self-adaptive systems, ensuring both their reliability and safety in complex operational environments.

Keywords— *self-adaptive systems; aviation; avionics; cybersecurity attacks; cybersecurity in avionics plug&fly avionics; aircraft security; zero-trust architecture; blockchains*

I. INTRODUCTION

The increasing size, complexity, availability requirements and dynamic environment of existing distributed software systems, derive the need of self-reliant and scalable tech solutions. Self-adaptive systems (SAS) are example of one such technology. In the area of avionics, SAS represents a huge potential for aircraft control and management, offering flexibility and resilience to changing operational conditions. By dynamically adjusting to environmental factors, system faults, and evolving mission requirements, self-adaptive avionics promise enhanced safety [1], maximum resource utilization, efficiency, and reliability [2] in aviation operations. However, the adaption techniques in SAS widen security attack surfaces by, i) exploiting adaptation strategy to use it against the system. For example, an attacker could collect a sufficient amount of information to reverse engineer the adaptation strategy in an attempt to deceive the system [3], or ii) exploiting multi-channel communication, sensors, devices, etc. and using those as attack surfaces [4]. Besides that, the feedback loops and control parameters can also be

maliciously fed to drive the adaptation decisions as per attackers' needs.

Since aviation and avionics industry has stringent security requirements as given in CS25-1319 and guided in DO-326¹ and DO-356²; these vulnerabilities could act as a bottleneck for adoption of the technology in avionics. While a few studies [5][6][7] discuss security of self-adaptive systems, they lack detailed analysis. Notably, no work has been done for analysis of security of self-adaptive systems for avionics yet, leaving a significant gap in literature. This study bridges this gap this study examines vulnerabilities in self-adaptive avionics and identifies potential security attacks that that exploit these weaknesses. For reliability, this study uses Plug&Fly Avionics (PAFA) [8], a self-adaptive avionics platform as a case study. PAFA is introduced to reduce the configuration and integration effort by the increasing complexity of integrated functions in Integrated Modular Avionics (IMA). Analysis of PAFA effectively aids in understanding the constructs of a self-adaptive system, and the impacts of self-adaptivity on the cybersecurity. Therefore, the major contributions of this study are following:

- Identification of security vulnerabilities introduced with SAS in avionics.
- Analysis of PAFA to understand the impact of identified security vulnerabilities.
- Proposal of novel mitigation strategies to overcome the challenges introduced by self-adaptivity.

The remaining paper is organized as follows: Section II discusses the related work in the field. Section III analyses the software architecture of self-adaptive system and highlights various cybersecurity challenges that are introduced with self-adaptivity. The PAFA use case further validates the authenticity of those findings by highlighting real application vulnerabilities. Section IV describes how the identified vulnerabilities are exploited to cause an attack. Section V describes novel mitigation strategies to ensure flexibility of self-adaptive systems is guaranteed without any additional security challenges. Finally, the paper concludes in Section VI.

II. RELATED WORK

Understanding the security landscape of self-adaptive avionics systems is important for design and development of secure systems complying to security requirements of RTCA DO-356 and DO-326 standards. These standards further

¹ <https://afuzion.com/do-326a-ed-202a-aviation-cyber-security/>

² <https://standards.globalspec.com/std/10398650/rta-do-356>

emphasize the need of rigorous testing, verification, and validation procedures to ensure the safety and integrity of airborne systems. SAS expose a dynamic attack surface as they adjust their configurations based on environmental changes. For instance, malicious environment changing itself can introduce new potential entry points for attackers, making it challenging to maintain a consistent and secure perimeter. Since the environment keeps changing, the security mechanisms must also respond accordingly. A comprehensive analysis of a written survey was conducted [9] in which 184 industry practitioners involved in development of SAS, responded. The findings of the analysis showed that multiple risks that arise during engineering a SAS include resource unavailability, misconfigurations in the system, implementation of adaptive security mechanisms, and system reliability issues. All of these either directly or indirectly lead to security breaches. Similarly, another study [10] discussed various types of adversarial attacks that can occur in MAPE-cycle of SAS. These attacks include spoofing, sensor manipulation, byzantine attacks, machine learning attacks, and many more. The study in [11] presented an adaptive neural network-based approach to mitigate the issues in distributed adaptive security consensus control. This was a study for a class of multi-agent systems under network decay and intermittent attacks. A downside of this study was that it did not consider non-ideal network topology. Another study [12] describes the importance of timely decision-making in SAS. The study indicates the adoption mechanisms introduce delays into control processes. It also suggests that these delays might be insignificant for some domains and critical for others. In terms of avionics, (depending on the use-case) timely decision making and execution of system functions has utmost importance to ensure safe and secure flight. Upon searching various online databases such as IEEE Xplore, ScienceDirect, and Google Scholar, only two studies [6][13] are found that present literature survey on security of SAS. But, these studies also have some limitations. The study in [6] presents a comprehensive survey for safety and security in SAS, examining the joint modeling of safety hazards and security threats. While it identifies security threats, it lacks in-depth analysis of their applicability. The study suggests further gap in safety and security analysis in SAS needs to be covered. Similarly, [13] presents security roadmap in terms of security patterns, which is a completely different area of study. Security patterns are architectures which are abstract and reusable frameworks, to solve a particular security problem. Our study, on the contrary presents a comprehensive review and analysis of potential cybersecurity attacks on self-adaptive avionics systems, while proposing that same design artifacts, security vulnerabilities and subsequent security attacks may also be prevalent to other domains.

III. IDENTIFICATION AND ANALYSIS OF SECURITY VULNERABILITIES IN SELF-ADAPTIVE SYSTEMS

To ensure that the features of self-adaptivity are utilized to its full potential, it is significant to identify and mitigate any accompanied security issues. This section firstly presents a review of cybersecurity vulnerabilities in self-adaptive systems. Secondly, the identified security challenges and vulnerabilities from this review are further rigorously analyzed with respect to PAFA. This analysis is important to understand the applicability and impact of identified vulnerabilities in avionics.

A. Potential Cybersecurity Vulnerabilities in Self-Adaptive Systems

Dynamic nature of SAS exposes the system to various new security vulnerabilities and exploitations by malicious actors [14]. The adaptation strategy implemented by a system has a potential to be used against it. Consider, the malicious entity collecting enough information from the system to reverse engineer its adaptation strategy, thereby deceiving the system. Consequently, the system operations are exploited. These attacks are unique to SAS [3]. Additionally, self-adaptive systems designed to be self-protecting, face uncertainties due to the specific factors influencing security, which can vary greatly by domain or implementation. For example, the requirements for timely adaptations in a self-driving vehicle differ significantly from those needed in a cloud-based server infrastructure [15]. This complicates effective designing due to unpredictable threats. Multi-channel communication in SAS for accommodate various devices, sensors, and other components widens open attack surfaces. Each communication channel, such as those between sensors, controllers, and external systems, can potentially be exploited if not properly secured. A very basic example would be the ability of an attacker to intercept the data transmitted between environmental sensors and the system's central processor, leading to incorrect and manipulated environmental calculations and potentially malicious adaptations based on tampered data. Feedback control loops in SAS, which are responsible for activities like collect, analyze, decide, and act, crucial for self-adaptive systems, can also be misused. For instance, an attacker could manipulate the data and feed incorrect information into the system. This leads to faulty and misguided decisions resulting in harmful adaptations. Furthermore, this type of exploitation compromises the ability of the system to effectively respond to real environmental threats or causing performance issues. Mapping of these security issues to various phases of the MAPE-K cycle is given in Table I. and illustrated in Fig. 1. This mapping enables the software engineers to develop targeted mitigation strategies for each phase to enhance system security and resilience.

B. Software Architecture of Plug&Fly Avionics (PAFA)

Plug&Fly Avionics (PAFA), is a self-configuring, self-optimizing, platform, which is also capable of hosting safety-critical functions in aircraft. The identification of security threats will improve the design and development process of PAFA and other self-adaptive aviation applications. The software architecture of PAFA (as taken from the publishing study) [8] is illustrated in Fig. 2.

TABLE I. MAPPING OF SELF-ADAPTIVE CYBERSECURITY ISSUES TO ITS MAPE-K CYCLE

Security Issue	MAPE-K Component
Exposure to new vulnerabilities due to dynamic nature	Knowledge (K), Plan (P)
Reverse engineering of adaptation strategy	Analyze (A)
Multi-channel communication leading to attack surfaces	Monitor (M)
Interception and manipulation of sensor data	Monitor (M)
Manipulation of feedback control loops	Monitor (M), Analyze (A), Knowledge (K)
Harmful adaptations due to manipulated data/plans	Plan (P), Execute (E)
Inability to respond to real environmental threats/performance issues	Analyze (A), Plan (P), Execute (E)

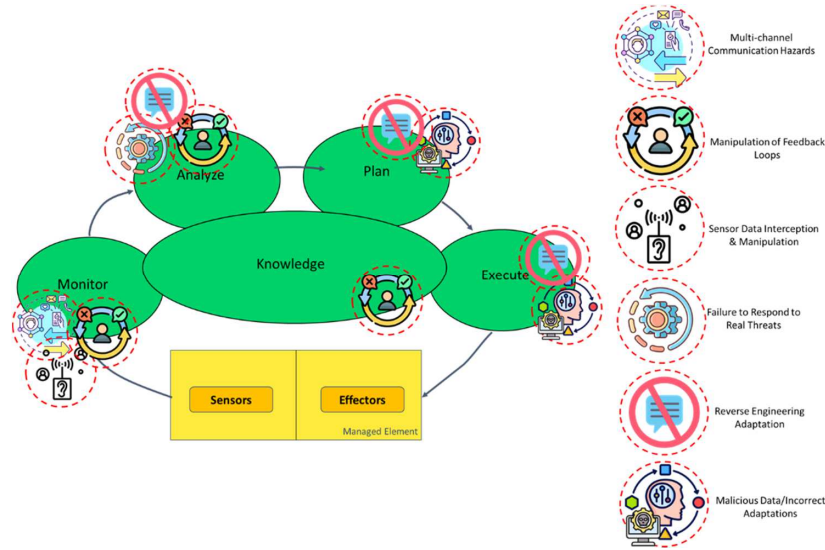


Fig. 1. Mapping of Self-Adaptive Cybersecurity Issues to MAPE-K Cycle

The SAS architecture is extended to MAP-QE-K, which comprises of monitor, analyze, plan, qualification, execution and knowledge. A PAFA module can host one or more out of this four partitions:

- The planning intelligence partition – for monitoring, analysis and planning. It also consists of “consciousness”, which is the knowledge counterpart, MAPK.
- The virtual qualification authority partition – for all qualification related tasks like safety- and security assessment and approval of the determined configuration Q.
- The verification execution unit – sub-component of qualification authority, serves as a sandboxed environment for testing a given schedule of tasks before execution Q.
- The life execution unit – for execution of predefined scheduled tasks during operation, E.

Cybersecurity vulnerabilities of PAFA architecture are discussed thoroughly in upcoming sub-sections. However, for deeper conceptual and functional information about these, refer to the previous studies [16][17] on PAFA.

1) Open Security Vulnerabilities in the PAFA Middleware

The middleware offers a unified interface for all configuration parameters in a PAFA module, making configuration management easier and facilitating application of coordinated reconfiguration within a module. However, PAFA’s dynamic reconfiguration introduces additional security concerns, such as tampering and manipulation attacks that exploit adaptation processes and compromise system integrity. PAFA operates on diverse hardware architectures, operating systems, and communication interfaces. While a consistent middleware API reduces reimplement effort it also raises interoperability and communication security challenges. Inconsistent protocols for communication in various devices are likely to result in loss of data integrity

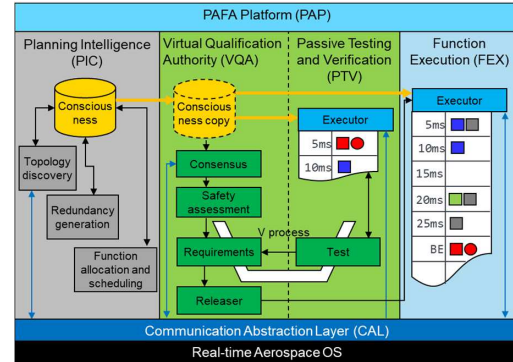


Fig. 2. PAFA Self-Adaptive Software Architecture

or unintended data exposure [18]. This means, that the attackers would get wide surface to conduct interception attacks. Exposing APIs for interoperability further increases the risk of injection attacks. The current ability of any node to request for task reconfigurations, leads to compromised nodes simultaneously sending reconfiguration requests to middleware beyond its capacity to process those, thereby flooding the system.

2) Open Security Vulnerabilities in the PAFA Planning Intelligence

The PAFA Planning Intelligence (PI) is responsible for decision-making in PAFA, specifically to enable finding a suitable configuration after first assembly of the platform as well as run-time adaptations of configuration. It utilizes the information from the knowledge base of PAFA, known as consciousness, to look for task configurations and scheduling. Besides consciousness, PI has three components, i.e., topology detection, redundancy generation and function allocation. These components, and their cybersecurity analysis are further as follows:

a) Consciousness

The information about different modules in PAFA is stored in the consciousness. It is a model-centric database to

manage PAFA's resources. The decision-making, task allocation and scheduling, and resource management and optimum utilization, of PAFA is highly reliant on the information contained by consciousness. Hence, if consciousness is compromised, the functionality of various PAFA components and devices will also be compromised. This means, any incorrect data that is fed into self-adaptive feedback loops can corrupt the information stored in consciousness. As a result, all the components accessing information will perform faulty actions. Slowing down of information retrieval from consciousness occur due to high loads of information retrieval [18], attackers can use this to deploy a Denial-of-Service (DoS) attack.

b) Topology Detection

The topology detection algorithm in PAFA collects the self-description information of connected nodes components and generates a model of the network topology from that information. In the topology detection protocol four types of device descriptive messages are specified. Any network device can mark its presence to the neighboring devices (network nodes) using these four message types. These include: i) advertise messages, are broadcasted periodically allowing identification of connected/disconnected neighbours, ii) an update message type to propagate new information through the network. iii) a self-description request and iv) self-description response message, which allow the nodes to collect information about each other. This self-description information can be requested by any node in the network, thereby giving a wide attack surface. For example, man-in-the-middle and other interception attacks can be deployed here to retrieve system information. This is also true

for any attacker that wants to inject/transmit outdated or incorrect information in the system [19]. Additionally, the intercepted data can be used to spoof a legitimate device and conduct malicious activities. Moreover, compromised nodes may not forward the network updates to the neighbours [20], impacting the destabilization of decisions based on network topology. Since topology discovery is important for generation of function redundancies, which is crucial for ensuring safety requirements, therefore any cyberattacks on topology discovery will impact the overall system.

c) Redundancy Generation

The redundancy generation or instantiation is a software component that analyses a simplex function specification and produces a redundant version, including redundancy mechanisms such as voting and monitoring. It does so by considering the detected hardware and maximal failure probability requirements specified for the system function. Redundancy generation cannot itself be accessed directly by an attacker. However, an impact on topology discovery and falsified self-discovery may have an impact on redundancy generation. The failure probability is provided by self-description, depicting that fraudulent self-description propagation, would result in a faulty redundancy instantiation, impacting the overall operation and safety of the aircraft.

d) Function Allocation

The function allocation software component takes the redundant application specification and the hardware model as input and computes and allocation of task blocks. Since the redundancy generation is driving force for this, any attacks applicable to that are also applicable to function allocation.

TABLE II. OPEN VULNERABILITIES IN SELF-ADAPTIVE AVIONICS CASE STUDY – PAFA

PAFA Unit	Sub-component (if applicable)	Vulnerabilities	Self-Adaptive Security Issues
Middleware	General	Inconsistent protocols for communication	Multi-channel communication leading to attack surfaces
		Ability of nodes to request reconfigurations at any given time	Exposure to new vulnerabilities due to dynamic nature
		Interoperability issues	Leads to adaptation strategy reverse-engineering
	Hypervisor/VM	Injecting undetectable malware in hypervisor OS	Reverse engineering of adaptation strategy
		Ability of infected VM to infect another VM	Harmful adaptations due to manipulated data/plans
		Data mobility	-
Planning Intelligence	Consciousness	Contains all critical information for PAFA operation and can be accessed using EOQ	Corrupted feedback leading to incorrect knowledge
		No backup maintained	Accumulation of incorrect data as there is no backup
		Entire functionality of PAFA platform depends on data in consciousness	Corrupted feedback leading to incorrect knowledge
	Topology Discovery	Simple self-discovery format, which is easy to exploit	Manipulation of feedback control loops
		Ability of nodes to request self-description from other node at any given time, without verification	Ability to intercept and manipulate data
		Responsibility of nodes to disseminate network information to neighbours	Inconsistent Feedback Leading to Faulty Analysis
Virtual Qualification Authority	Redundancy Generation	Compromised node may propagate false self-description, generating false redundancies	Compromised feedback loops affecting knowledge
	Function Allocation	Depends on redundancies, that can be incorrect	Harmful adaptations due to manipulated data/plans
	Verification Execution	-	
	Consensus	Dependency on nodes for decision making, which may be compromised	Reverse engineering of adaptation strategy
Live Execution	-	Resource exhaustion	Failure to respond to real threats
		Hardware dependencies/failures at run time	Inconsistent execution
		Verified reconfigurations not reaching this unit	Harmful adaptations due to manipulated data/plans

3) Open Security Vulnerabilities in the PAFA Virtual Qualification Authority (VQA) and Verification Execution (VE)

VQA is responsible to verify new schedule of tasks and new routing tables before these are active. The VQA acts as a gatekeeper for the VE unit which executes all the configurations during operation. The VQA receives a copy of the consciousness to be able to verify task allocation for execution unit. Therefore, if the consciousness is compromised, the decisions by VQA will also be compromised. The VQA incorporates consensus protocols, and based on the protocol used, the risk of certain security threats like Sybil attacks and 51% attack opens up. This reliance on consensus protocols illustrates a common vulnerability in self-adaptive systems, i.e., one general assumption is that the decision lies with nodes in consensus, therefore if majority of the nodes are compromised, the decisions can be drove according to the adversary's wishes.

4) Open Security Vulnerabilities in the PAFA Live Execution Unit (LEU)

Since the execution of tasks in real time takes place in this unit, hence it holds a great importance with respect to security. Several security vulnerabilities are likely to arise here, including the ability of an attacker to exhaust resources. These resources include memory, computing power, bandwidth and so on. An attacker can overload a system resource by sending excessive requests, ensuring the unavailability of the resource to legitimate requests, causing a DoS. Since PAFA is a SAS, it is greatly influenced by the surrounding environment, which makes it prone to this vulnerability.

5) Open Security Vulnerabilities in the PAFA Virtualization

The four units of the PAFA software architecture operate concurrently in distinct virtual machines (VMs). While this setup offers security isolation in accordance with the DO-356 standard, it also introduces risks associated with virtualization [21] and hypervisor security [22]. The increased complexity with virtualization likely creates larger attack surfaces and greater opportunities for exploitation. The hypervisor manages physical hardware resources for virtualization within PAFA, making it a prime target for attackers and data exfiltration attacks. It has full access to resources, data, VMs, memory, and I/O. Additionally, the hypervisor software may be susceptible to buffer overflow vulnerabilities, which can enable an attacker to send harmful inputs that overwrite adjacent memory locations, potentially executing arbitrary code. These vulnerabilities in the hypervisor underscore the significant challenges that self-adaptive systems face, as the dynamic resource management needed for adaptability can unintentionally expose critical components to sophisticated attacks. The vulnerability and attack analysis of PAFA is summarized in Table II.

IV. ANALYSIS OF CYBERSECURITY ATTACKS TO PLUG&FLY AVIONICS (PAFA) – A SELF-ADAPTIVE USE CASE

This section describes the attacks, exploits and impacts of previously discussed vulnerabilities as discussed.

A. Injection Attacks

SQL-injection-like attacks are applicable on PAFA. Consciousness is accessible by using EOQ, which is derived from OCL comparable to SQL. Hence the applicability of

SQL injection like attacks lies here. This potentially impacts decision making in PAFA. The resulted decisions may cause system failures and faulty operations. However, the probability of this attack is extremely low as the EOQ interface is not open for access to public.

B. Denial of Service (DoS) and Distributed Denial of Service (DDoS attacks)

DoS and DDoS are among the most crucial attacks in for PAFA since the concept involves using shared hardware and software resources. These attacks are applicable at various units in PAFA platform including topology discovery, function allocation, and middleware communication. Since a number of vulnerabilities in various PAFA modules can induce this type of attacks, the risk of these attacks is high in comparison to others.

C. Man-in-the-Middle Attacks

Tapping the communication lines in PAFA is likely to help the attacker gain valuable insights to the network regarding various components attached and the data packet structure. This information can be used to inject false data using the same structure, or impersonate a particular node to deceive the neighbours. The data gathered through this attack can also be used for side channel attacks impacting PAFA decisions.

D. Spoofing Attacks

The attacker may add false data into the system, resulting in incorrect decisions [23]. The attacker can fabricate its self-description message to follow the pre-defined structure and act like a legitimate device attached to the network. With these attacks, attackers i) provide inaccurate information regarding connectivity and relationships among network nodes, ii) manipulate network routing by injecting false information about the routing protocols in the network. This leads to incorrect selection of paths for transmission of the data, and may also compromise the performance of the system by redirecting all the data at the same path. This is known as topology spoofing. Besides that, PAFA platform uses multiple sensors including altitude sensors, sensors for control input, etc. If the input data for these sensors is spoofed the PAFA system functions will be populated with incorrect or falsified facts. Furthermore, the attacker may craft messages and broadcasts them to neighbours in a way that these appear to come from the entity whose identity is spoofed. Using this attack, the attacker may request unneeded resources and reconfigurations to exhaust the system.

E. Virtualization Attacks

As mentioned earlier, hypervisor in virtualization is a profitable target for attackers. Hence relevant mitigations have to be applied to protect PAFA platform from such attacks. However, initiating a data theft attack on it is not easy in a controlled and isolated environment that PAFA or other self-adaptive avionics offer.

The findings of this section are summarized in Table III which describes various attack scenarios specific to PAFA to show how each of the given vulnerabilities can be exploited to execute cyberattacks.

V. MITIGATION STRATEGIES FOR SECURITY ATTACKS ON SELF-ADAPTIVE AVIONICS

This section proposes Zero Trust, Blockchain, and multilayered security as solutions to self-adaptive cybersecurity challenges.

TABLE III. EXPLOITED VULNERABILITIES AND ATTACK SCENARIOS

Attack	Attack Scenario	Exploited Vulnerability	Impact
Malware and Viruses	Injecting malware in Hypervisor	Hypervisor OS	System malfunctioning
Query Language Injection	Having sound knowledge of EOQ, incorrect data is fed into PAFA system.	Lack on restriction of entities that can access consciousness using EOQ	Incorrect adaptation decisions.
Denial of Service Attacks	Multiple reconfiguration requests to middleware beyond its processing capacity	Ability of any node to request for reconfiguration of task allocation	Disruption of task allocation and execution
	Multiple self-description messages to nodes, beyond their processing capacity	Any node can request self-description message from any other node	Nodes are overwhelmed and unresponsive due to multiple messages/requests from malicious nodes
	Multiple EOQ requests to consciousness beyond its processing capacity	Ability of data retrieval from consciousness using EOQ	Consciousness is unresponsive, PI and VQA do not receive required data for execution, halting the system
Man-in-the-Middle	Tapping information flow structure, injecting similarly structured falsified data	Simplistic data structure	Mislead flight system → unsafe flight and unpredictable platform behavior.
Topology Spoofing	Malicious device mimics self-description message and provide false topology information.	Simplistic self-description data structure. Ability for any node to request self-description	Reliance on network nodes that probably do not even exist.
	Injecting false advertisement messages, implying the presence of nodes, which do not exist.	Ability of nodes to advertise themselves to neighbours as desired	Reliance on network nodes that probably do not even exist.
	Node not forwarding the neighbor topology information received.	Ability of nodes to send/request messages neighbours as desired	Incomplete topology information, incorrect decisions
Sensor Spoofing	Malicious entity mimics PAFA sensors such as altitude sensor, temperature sensor and produces false readings for PAFA	Sensor data manipulation	Disruption of control laws that govern cyberphysical behaviour of PAFA
ID Spoofing	Misusing ID of another node to generate false reconfiguration requests	Lack of secure authentication techniques.	Incorrect or delayed resource allocation and reconfigurations
	Attacker sends a malicious message with a spoofed identifier, mimicking legitimate PAFA communication.	Reliance of message routing only on identifier	Generation of unauthorized reconfigurations requests to VQA. Inability to identify attack origin
Virtualization Attacks	Attacker exploits buffer overflow to access flight controls and execute arbitrary command	Buffer overflow lets attackers overwrite memory and execute malicious code via crafted input.	Significant risks to flight safety and integrity.
51% Attack	Attacker holds 51% of the network and drive system decisions to his wishes	Consensus mechanisms	Erroneous flight data, communication disruptions, or even loss of control over the aircraft.

A. Zero-Trust Architecture (ZTA)

In a self-adaptive system, such as PAFA, where dynamic reconfigurations and multi-channel communication amplify attack surfaces, ZTA mitigates vulnerabilities by enforcing granular, adaptive security controls. Using Identity and Access Management (IAM), it ensures only verified entities interact with critical components like middleware or consciousness, preventing unauthorized access. Similarly, the feature of network segmentation in ZTA isolates feedback loops and subsystem communications, limiting the impact of malware injection or topology spoofing. Further, node integrity is evaluated before granting participation in reconfiguration processes, preventing manipulated redundancies or virtual qualifications. Data security further ensures the integrity of sensor data and blockchain-inspired tamper-evident logs. Lastly, continuous monitoring and analytics in ZTA proactively identify anomalous behaviors in adaptation requests or feedback loops, countering potential attacks like ID spoofing and topology manipulation.

B. Blockchain Networks

Blockchain immutable ledger uses hashing techniques such as SHA-256, which securely records environmental sensor data and adaptation actions, ensuring each entry is verifiable. This guarantees that decisions are based on unaltered data, crucial for maintaining operational integrity. Blockchain, moreover, ensures decentralized identity management by assigning Decentralized Identifiers (DIDs) to

devices, validated through protocols like Elliptic Curve Cryptography (ECC), ensuring that only authenticated devices can access the system, effectively blocking spoofing attempts. Smart contracts automate the process of access control and adaptation protocols, thereby enforcing strict rules without a central authority preventing malicious reconfiguration and ensuring legitimate adaptations. Lastly, blockchain's transparent auditability provides an unchangeable record of all transactions, utilizing cryptographic proofs to enable quick identification of anomalies and forensic analysis. Together, these components create a secure, adaptive environment.

C. Comprehensive Multilayered Security Frameworks

Given PAFA's hierarchical structure, a multilayer security architecture [24] could be promising solution. The security measures can be applied on physical, networks and application layers. Physical layer security counters physical tampering with sensors, flight control modules, and communication devices. Network layer security, restricts malicious lateral movements by employing firewalls, Intrusion Detection and Prevention Systems (IDS/IPS), and Secure Network Segmentation (SDNs). Additionally, Application layer security integrates safe coding practices, and runtime fault-handling mechanisms for protection against malicious exploitation of dynamic reconfigurations. should be followed in avionics software that dynamical work. This layered strategy minimizes the risk of cascading failures from vulnerabilities like middleware inconsistencies or hypervisor attacks.

VI. CONCLUSION

This paper provides detailed analysis on current cybersecurity landscape focusing on self-adaptive avionics. Plug&Fly Avionics (PAFA) serves as a case study, highlighting numerous vulnerabilities introduced by self-adaptivity that threaten the integrity and reliability of critical avionics operations. Some conclusions derived from the findings of this study are: i) The identified cybersecurity vulnerabilities include inconsistent communication protocols for multi-channel communication, compromising of feedback loops, lack of data backup, intercepting and/or manipulating sensor data that the system decisions rely on, and so on. ii) Exploiting the vulnerabilities mostly result in faulty decisions and/or operational failures which can be catastrophic. iii) Typical security solutions are inadequate in such new challenges. Thus, new and creative solutions must be offered to overcome the destruction. iv) Latest technologies like Blockchains, Zero-Trust architecture and multilayered security emerge as promising solutions to existing challenges. This work contributes towards filling in the existing gaps concerning self-adaptivity in avionics and lays the groundwork for further research in this area. Using these findings, more resilient, robust and dynamic cybersecurity solutions can be designed in future to overcome newly introduced challenges.

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